

1.

Spin Inertia : $700 \text{ kg} \cdot \text{m}^2$

$$F = 2.5 \text{ N}$$

$$R_s = 1 \text{ m}, R_A = 1 \text{ m}$$

$$\Delta t = 0.25 \text{ sec}$$

$$I_{sp} = 200 \text{ sec}$$

 F, R, θ given $\rightarrow \frac{\Delta \phi}{\Delta t} \text{ cal mp}$

(a)

$$\vec{M} = \frac{d\vec{H}}{dt} = \vec{R} \times \vec{F}$$

$$RF = C \cdot \frac{d\omega}{dt} = \frac{\Delta \omega}{\Delta t}$$

$$\Delta t = \frac{\Delta \omega}{RF} \quad (\text{pure spin})$$

(a)

$$\frac{\Delta \omega}{\Delta t} = 60 = 24$$

$$I = Fat = 2.5$$

$$\omega = \frac{I}{I_{sp}} \quad \therefore I_{sp} = \frac{I}{\omega} = \frac{I}{m_p \cdot g_0}$$

$$\therefore m_p = \frac{I}{I_{sp} \cdot g_0}$$

$$I_{sp} = \frac{V_e}{g} = \frac{V_e \cdot dm}{g \cdot dm} = \frac{dV \cdot m \times \frac{dt}{dm}}{g \cdot dm} = \frac{m \frac{dV}{dt} \cdot dt}{g \cdot dm}$$

$$= \frac{I}{\omega} = \frac{I}{m_p \cdot g_0}$$

$$m_p = \frac{I}{I_{sp} g_0}$$

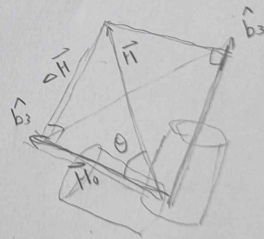
(b)

$$\vec{R} \times \vec{F} = \vec{M} = \frac{d\vec{H}}{dt} = \frac{\Delta \vec{H}}{\Delta t}$$

$$\therefore \Delta \vec{H} = (\vec{R} \times \vec{F}) \Delta t$$

$$\Delta t = \frac{|\Delta \vec{H}|}{|\vec{R} \times \vec{F}|}$$

$$\begin{aligned} \Delta H_{tot} &= 2 |\Delta \vec{H}| \\ &= 2 |\vec{H}_0| \cdot \tan 50 \end{aligned}$$



2.

$$\vec{B} = R \cdot \vec{N}$$

$$\vec{B} = \begin{bmatrix} \cos 20 \cdot \cos 20 - \sin 20 \cdot \cos 20 \cdot \sin 20 & \sin 20 \cdot \cos 20 + \cos 20 \cdot \cos 20 \cdot \sin 20 & \sin 20 \cdot \sin 20 \\ -\cos 20 \cdot \sin 20 - \sin 20 \cdot \cos 20 \cdot \cos 20 & -\sin 20 \cdot \sin 20 + \cos 20 \cdot \cos 20 \cdot \cos 20 & \sin 20 \cdot \cos 20 \\ \sin 20 \cdot \sin 20 & -\cos 20 \cdot \sin 20 & \cos 20 \end{bmatrix}$$

$$\vec{B} = \begin{bmatrix} -0.1736 & 0.5246 & 0.8335 \\ -0.5246 & -0.7655 & 0.3726 \\ 0.8335 & -0.3726 & 0.4081 \end{bmatrix} \cdot \vec{N}$$

↳ rotation matrix

in principal Rotation theorem,

$$\sin \varphi = \frac{1}{2} [\lambda_3 (c_{12} - c_{21}) + \lambda_2 (c_{31} - c_{13}) + \lambda_1 (c_{23} - c_{32})]$$

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} \text{RotMat} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix}$$

$$\{\hat{b}\} = [c] \{\hat{n}\}$$

$$\hookrightarrow \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = [c] \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} \rightarrow [c] \hat{\lambda} = \hat{\lambda} \quad \text{eigenvalue} \rightarrow \varphi$$

3.

$$\vec{T} = \dot{\vec{H}} + \vec{\omega} \times \vec{H}$$

$$\vec{H} = I_1 \omega_1 \hat{e}_1 + I_2 \omega_2 \hat{e}_2 + I_3 \omega_3 \hat{e}_3 + h_2 \hat{e}_2$$

$$\vec{\omega} \times \vec{H} = \begin{vmatrix} \hat{e}_1 & \hat{e}_2 & \hat{e}_3 \\ \omega_1 & \omega_2 & \omega_3 \\ I_1 \omega_1 & I_2 \omega_2 + h_2 & I_3 \omega_3 \end{vmatrix}$$

$$= [(I_3 - I_2) \omega_2 \omega_3 - h_2 \omega_3] \hat{e}_1 + (I_1 - I_3) \omega_1 \omega_3 \hat{e}_2 + [(I_2 - I_1) \omega_1 \omega_2 + h_2 \omega_1] \hat{e}_3$$

$$\dot{\vec{H}} = I_1 \dot{\omega}_1 \hat{e}_1 + I_2 \dot{\omega}_2 \hat{e}_2 + I_3 \dot{\omega}_3 \hat{e}_3 + h_2 \dot{\hat{e}}_2$$

$$T_1 = I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 - h_2 \omega_3$$

$$T_2 = I_2 \dot{\omega}_2 + h_2 + (I_1 - I_3) \omega_1 \omega_3$$

$$T_3 = I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 + h_2 \omega_1$$

$$\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = R_3(\psi) R_2(\theta) R_1(\phi) \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

4.

active pitch

$$I_2 \ddot{\theta} + \dot{h}_2 = T_2$$

$$\dot{h}_2 = K_p (\theta + z_p \dot{\theta})$$

$$I_2 \ddot{\theta} + K_p z_p \dot{\theta} + K_p \theta = T_2$$

$$\frac{\theta(s)}{T_2(s)} = \frac{1}{I_2 s^2 + K_p z_p s + K_p}$$

$$\ddot{\theta} + \frac{K_p z_p}{I_2} \dot{\theta} + \frac{K_p}{I_2} \theta = \frac{T_2}{I_2} = F_2$$

$$\ddot{\theta} + 2 \frac{z_p K_p}{2 I_2} \dot{\theta} + \omega_p^2 \theta = F_2$$

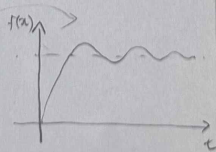
$$\theta(s) = \frac{T_2(s)}{I_2 (s^2 + 2 \xi_p \omega_p s + \omega_p^2)}$$

$$\lim_{s \rightarrow 0} \theta(s) = s \frac{T_2}{I_2 s^2 + K_p z_p s + K_p} \frac{1}{s} = \frac{T_2}{K_p}$$

$$\theta_{ss} = \frac{T_2}{K_p}$$

$$\therefore K_p = \frac{T_2}{\theta_{ss}}, \quad z_p = \frac{2 \xi_p}{\sqrt{\frac{K_p}{I_2}}}$$

$$\ddot{x} + 2\beta \dot{x} + \gamma^2 x = 0$$

i) underdamping : $\gamma^2 > \beta^2$ ii) overdamping : $\gamma^2 < \beta^2$ iii) critical damping : $\gamma^2 = \beta^2$ 

$$\text{roll: } I_1 \ddot{\phi} - h_2 \dot{\psi} + \omega_0 h_2 \phi = T_{G5} \alpha + T_{S1}$$

$$\text{yaw: } I_3 \ddot{\psi} + h_2 \dot{\phi} + \omega_0 h_2 \psi = T_{S4} \alpha + T_{S3}$$

$$T = -K(\phi + z \dot{\phi})$$

$$I_1 \ddot{\phi} + K_L \cos \alpha \dot{\phi} + (\omega_0 h_2 + K \cos \alpha) \phi - h_2 \dot{\psi} = T_{S1}$$

$$I_3 \ddot{\psi} + \omega_0 h_2 \psi + (K \sin \alpha + h_2) \dot{\phi} + K \sin \alpha \phi = T_{S3}$$

$$(\omega_0 h_2 + K \cos \alpha) \phi_{ss} = T_{S1}$$

$$\phi_{ss} = \frac{T_{S1}}{\omega_0 h_2 + K \cos \alpha}$$

$$\omega_0 h_2 \psi_{ss} + K \sin \alpha \phi_{ss} = T_{S3}$$

$$\psi_{ss} = \frac{T_{S3} - K \sin \alpha \phi_{ss}}{\omega_0 h_2}$$

$$h_2 \dot{\psi} = -T_{G5} \alpha$$

$$\Rightarrow T = -\frac{h_2 \dot{\psi}}{\cos \alpha}$$

$$I_3 \ddot{\psi} + \omega_0 h_2 \psi = T_{S4} \alpha + T_{S3}$$

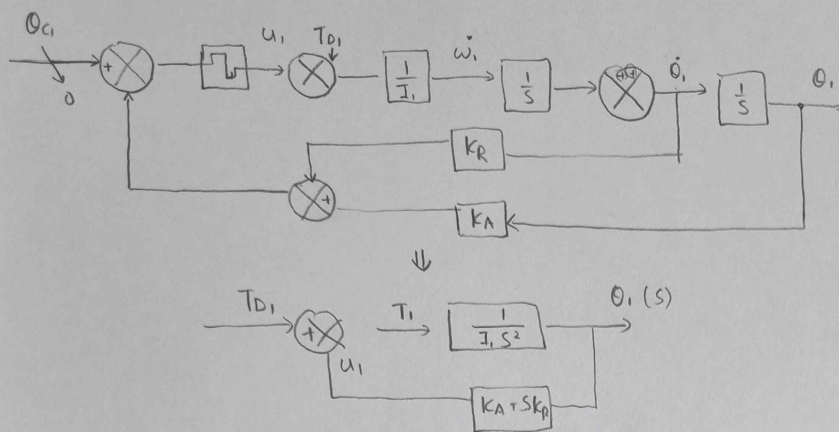
$$= -\left(\frac{h_2 \dot{\psi}}{\cos \alpha}\right) \sin \alpha + T_{S3}$$

$$I_3 \ddot{\psi} + h_2 \dot{\psi} \tan \alpha + \omega_0 h_2 \psi = T_{S3}$$

$$\psi_{ss} = \frac{T_{S3}}{\omega_0 h_2}$$

$$h_2 = \frac{T_{S3}}{\omega_0 \psi_{ss}}$$

$$K = \frac{T_{S1}}{\cos \alpha \phi_{ss}} - \frac{\omega_0 h_2}{\cos \alpha}$$



closed loop TF : $\frac{\theta_o}{\theta_i} = T.F$

$\theta_o = T.F \theta_i$

